UI Prototypes Specification

for

VisionEdge: AI-Powered Optical Store Management & Virtual Try-On System

Version 1.0

Design System: Bootstrap 5.3 + Custom CSS Variables + Dark/Light Mode

Interactions: MediaPipe Computer Vision, Canvas API, AOS, Swiper

Date: August 2026

Original UI Prototypes & Interface Design Document

for

VisionEdge: AI-Powered Optical Store Management & Virtual Try-On Syste

Version: 1.0

Design System: Bootstrap 5.3 + Custom CSS Variables + Dark/Light Theme Switching

Interactions: MediaPipe Computer Vision, Canvas API, AOS Animations, Swiper Carousel

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1. Design System & UI Architecture

The VisionEdge user interface is built on a responsive Bootstrap 5.3 foundation augmented with modern CSS variable design tokens, glassmorphism cards, and dynamic light/dark theme switching stored in local storage (data-bs-theme).

Core Design Tokens & Palette:

```
:root {
```

```
    --primary-color: #2563eb;      /* Royal Blue (Primary Action/Buttons) */
    --secondary-color: #0f172a;    /* Slate Navy (Headers & Hero Overlay) */
    --accent-color: #06b6d4;       /* Neon Cyan (AI Highlights & Badges) */
    --success-color: #10b981;      /* Emerald Green (Active/Completed Status) */
    --warning-color: #f59e0b;      /* Amber (Pending Alerts & Low Stock) */
    --card-bg: #ffffff;            /* Card Container Background */
    --text-muted: #64748b;         /* Muted Subtitles & Labels */
    --border-radius-lg: 16px;      /* Rounded Large Card Radius */
    --shadow-sm: 0 1px 3px rgba(0, 0, 0, 0.1);
    --shadow-lg: 0 10px 25px -5px rgba(0, 0, 0, 0.1);
}
```

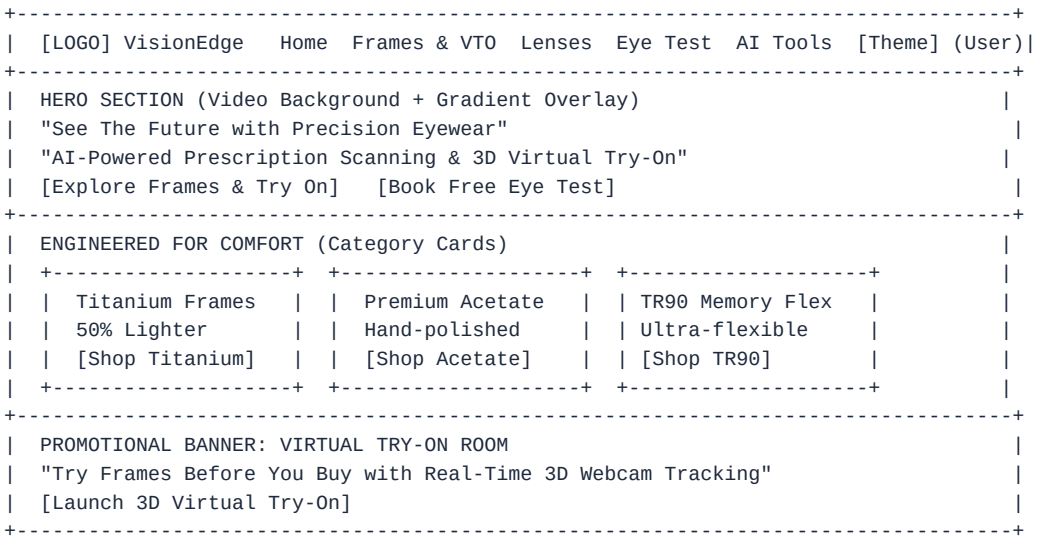
Global Shared Components:

1. Top Navigation Bar (navbar-container):
   * Brand Logo: VisionEdge Opticals with icon .
   * Main Nav Links: Home, Frames & VTO, Lenses, Eye Test, AI Recommendations, About.
   * Utility Actions: Light/Dark Theme Switcher toggle, Shopping Cart badge, Wishlist counter, User Profile avatar.
2. Global Preloader (#loader):
   * Centered CSS spinner with smooth fade-out animation upon page mount.
3. Footer (footer-container):
   * 4-column layout: Store Branding, Quick Links, Optical Care Services, Store Location & Contact Details.

2. Page Layout Prototype Specifications

2.1 Prototype 1: Homepage & Customer Landing (`index.html`)

The entry point for retail customers showcasing eyewear collections and computer vision AR features.

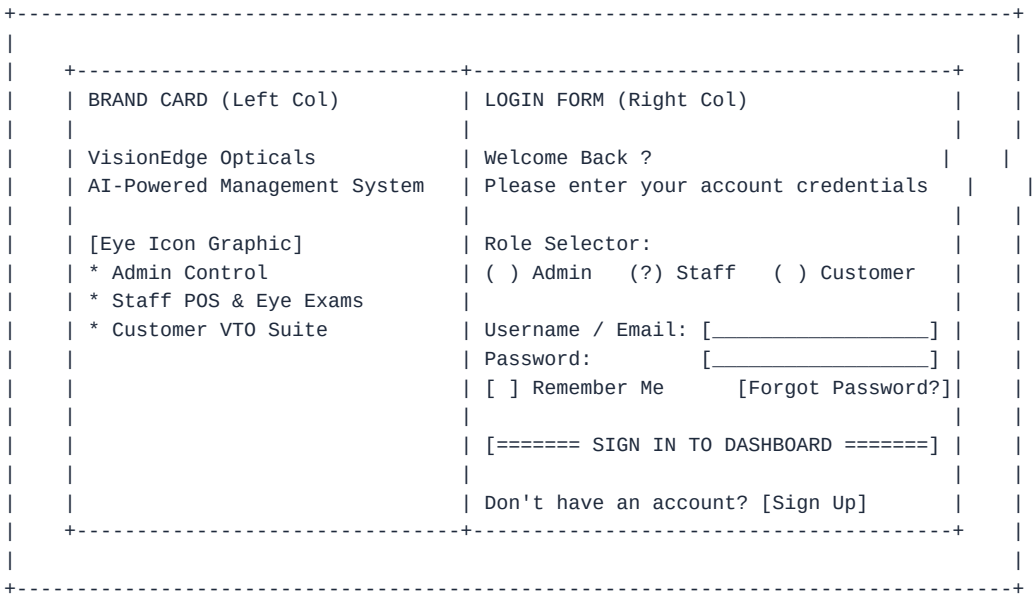


Key UI Interactions:

- Video background player (hero-video) with dark linear gradient overlay (hero-overlay).
- Dynamic text animation powered by Typed.js highlighting optical benefits.
- Category cards with hover lift animations (transform: translateY(-5px)).

2.2 Prototype 2: Authentication Suite (`login.html` & `signup.html`)

Provides secure JWT access for Store Administrators, Optometrists/Staff, and Customers.

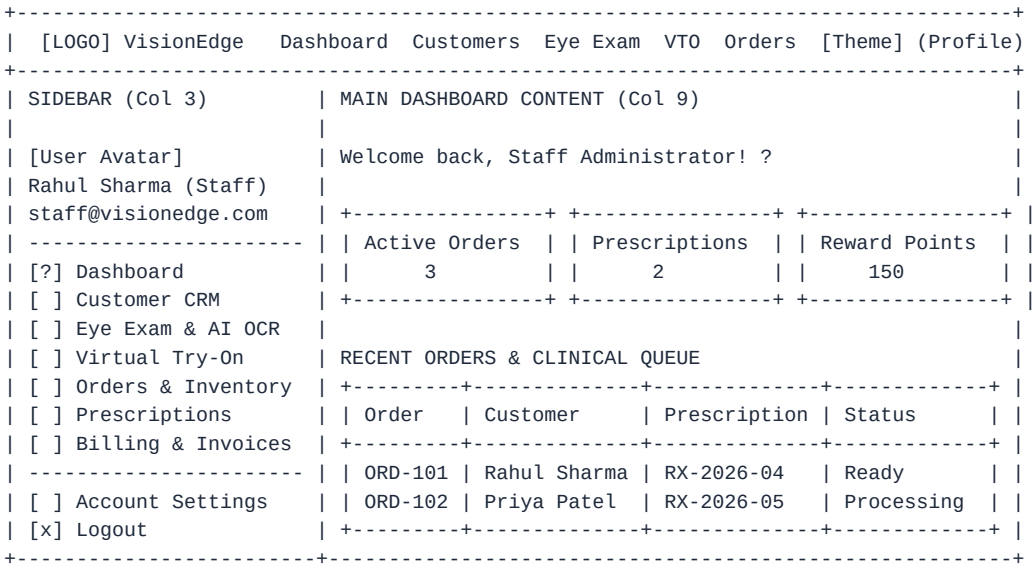


Key UI Interactions:

- Role selector radio pills allowing instant credential routing.
- Toggle password visibility eye icon.
- Form validation with interactive error popups.

2.3 Prototype 3: Staff & Customer Dashboard (`dashboard.html`)

Central workspace for managing clinical tasks, orders, saved prescriptions, and store metrics.

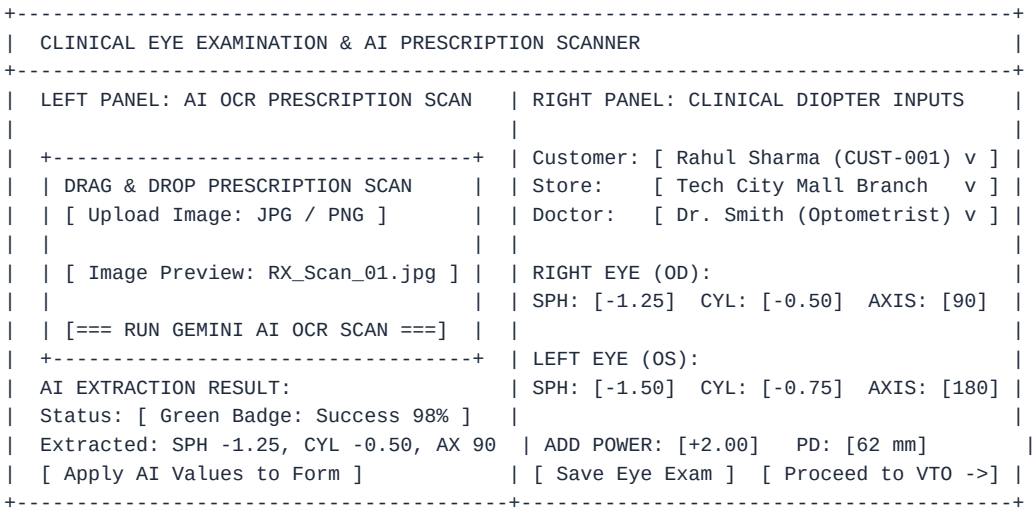


Key UI Interactions:

- Sticky sidebar with active tab navigation states (.nav-link.active).
- KPI stat cards displaying numeric counters with background opacity icons.

2.4 Prototype 4: Clinical Eye Exam & AI OCR Scanner (`eye-test.html`)

Enables optometrists to record clinical diopters or scan physical prescription scans using Google Gemini AI.

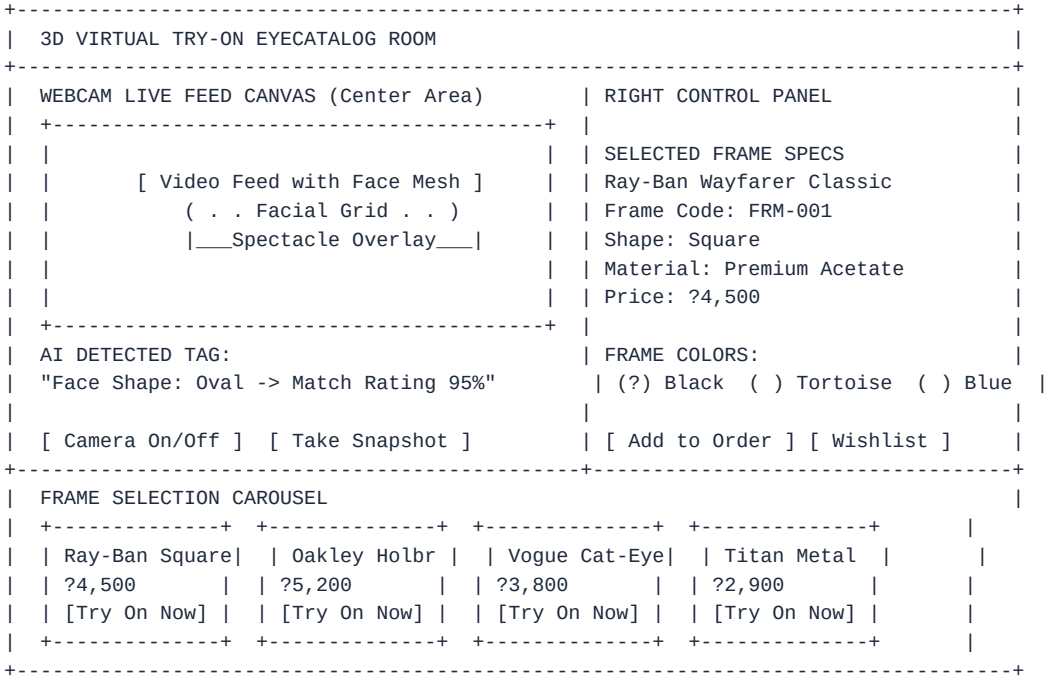


Key UI Interactions:

- Drag-and-drop file upload container with instant canvas preview.
- Progress spinner during Gemini API processing.
- Auto-fill button transferring parsed JSON values into form inputs.

2.5 Prototype 5: Virtual Try-On Suite & Frame Catalog (`frames.html` & `products.html`)

Interactive browser room where customers preview spectacle frames on their face via webcam.



Key UI Interactions:

- HTML5 MediaPipe face mesh canvas video stream.
- Frame carousel switcher updating the active overlay PNG in real time.

- Canvas snapshot generator producing downloadable try-on photos.

2.6 Prototype 6: AI Lens Recommendation Engine (`lens-recommendation.html`)

Suggests optimal lens designs and coatings based on prescription diopters.

```
+-------------------------------------------------------------------------------+
|  AI OPTICAL LENS RECOMMENDATION ENGINE                                        |
+-------------------------------------------------------------------------------+
|  PRESCRIPTION INPUT PARAMETERS                | AI LENS RECOMMENDATION RESULTS  |
|  Customer: Rahul Sharma                       |                                 |
|  Prescription: SPH -2.50, CYL -0.75, ADD +2.00 | Recommended Design:            |
|  Usage: 8+ Hours Screen Time (Digital Devices) | [ Badge: Progressive Lens ]    |
|                                               | High near-addition power (+2.00)  |
|  SELECT LENS COATINGS:                        | requires seamless multifocal.   |
|  [x] Anti-Reflective (Crizal)                 |                                 |
|  [x] Blue Cut (Digital Protection)            | Recommended Coating:            |
|  [ ] Photochromic (Sun-Adaptive)              | [ Badge: Zeiss BlueGuard ]      |
|                                               | Blocks harmful 420nm blue light. |
|  [=== GENERATE AI RECOMMENDATION ===]         | Price: ?3,500                   |
|                                               | [ Select Lens for Order ]       |
+-----------------------------------------------+---------------------------------+
```

2.7 Prototype 7: Checkout & POS Invoicing (`checkout.html`)

Point-of-Sale checkout interface calculating items, GST taxes, and issuing invoices.

```
+-------------------------------------------------------------------------------+
|  POINT OF SALE CHECKOUT & BILLING                                             |
+-------------------------------------------------------------------------------+
|  ITEMIZED ORDER SUMMARY (Left Col 7)          | BILLING & TAX BREAKDOWN (Col 5) |
|  +-----------------------------------------+  |                                 |
|  | Item                  Qty   Price Total |  | Subtotal:            ?8,000.00 |
|  +-----------------------------------------+  | Discount:             - ?500.00 |
|  | Ray-Ban Wayfarer Frame   1   ?4,500 ?4,500 |  | Taxable Amount:      ?7,500.00 |
|  | Zeiss BlueGuard Lens     1   ?3,500 ?3,500 |  | GST Tax (18%):       ?1,350.00 |
|  +-----------------------------------------+  | ------------------------------- |
|                                               | NET PAYABLE TOTAL:    ?8,850.00 |
|  CUSTOMER DETAILED INFO:                      |                                 |
|  Name:  Rahul Sharma  (CUST-001)              | PAYMENT TENDER METHOD:          |
|  Phone: 9812345678                            | (?) Cash   ( ) Card   ( ) UPI   |
|  City:  Mumbai                                |                                 |
|                                               | [=== COMPLETE BILL & PRINT ===] |
|  [ + Add Frame Item ]   [ + Add Lens Item ]   | [ Download PDF Invoice ]        |
+-----------------------------------------------+---------------------------------+
```

2.8 Prototype 8: Prescriptions, Orders & Payment History (`prescriptions.html`, `orders.html`, `paym

Customer history portal for viewing past medical eye tests and order receipts.

```
+-------------------------------------------------------------------------------+
|  CUSTOMER SAVED PRESCRIPTIONS & ORDER HISTORY                                 |
+-------------------------------------------------------------------------------+
|  PRESCRIPTION HISTORY (Card View)             | SALES ORDERS HISTORY            |
|  +-----------------------------------------+  | +---------------------------+ |
|  | Prescription #: RX-2026-04              |  | | Order #: ORD-1001         | |
|  | Date: 15 Aug 2026 | Dr. Smith            |  | | Date: 15 Aug 2026         | |
|  | OD: SPH -1.25  CYL -0.50  AX 90         |  | | Total: ?8,850.00          | |
```

```
|  | OS: SPH -1.50  CYL -0.75  AX 180          |  | | Status: [ Green: Delivered] | |
|  | [ View Scan Image ]  [ Download PDF ]     |  | | [ Download PDF Invoice ]    | |
|  +-------------------------------------------+  | +-----------------------------+ |
+-------------------------------------------------+---------------------------------+
```

3. Component Design System & Responsive Breakpoints

Target Breakpoints:

- Desktop Grid (>= 1200px): Full multi-column view with sidebar navigation.
- Tablet Grid (768px - 1199px): Collapsible sidebar drawer, 2-column card layouts.
- Mobile Web View (< 768px): Stacked single-column layouts with off-canvas slide-out navbar menu.

End of UI Prototypes & Interface Design Document for VisionEdge.